BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 2 Practical No: 3

Roll no: 37 PRN No: 12412175 Name of Student: Rugved Raut

Subject Name : Deep Learning

Problem Statement

Implement Forward Propagation and Backpropagation using TensorFlow/Keras. Analyze the effect of different learning rates and the number of epochs on model performance.

Algorithm

1. Start the program.
2. Import the required libraries: TensorFlow, Keras, Matplotlib.
3. Load the MNIST handwritten digit dataset using mnist.load_data().
4. Normalize the training and testing images by dividing pixel values by 255.
5. Create a Sequential Neural Network model.
6. Add a Flatten layer to convert 28×28 images into a 1D vector.
7. Add a hidden Dense layer with 128 neurons using ReLU activation.
8. Add an output Dense layer with 10 neurons using Softmax activation.
9. Compile the model using the Adam optimizer with a learning rate of 0.001 and Sparse Categorical Crossentropy loss function.
10. Train the model for 5 epochs using the training dataset.
11. Plot the training and validation accuracy graph.
12. Plot the training and validation loss graph.
13. Evaluate the trained model using the testing dataset.
14. Display the testing loss and testing accuracy.

15. Train the model using different learning rates (0.1, 0.01, 0.001, 0.0001) and compare the accuracies.
16. Train the model using different epochs (1, 5, 10, 20) and compare the accuracies.
17. Analyze the effect of learning rates and epochs on model performance.
18. Stop the program.

Code

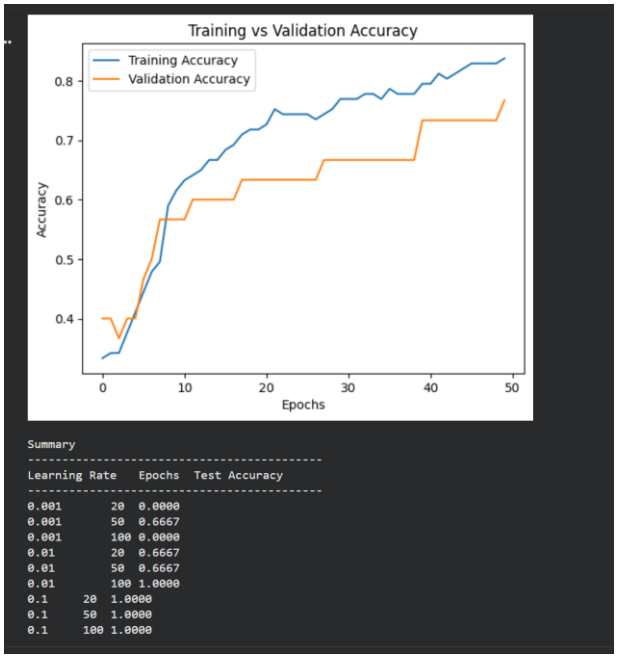
Google Colab:

https://colab.research.google.com/drive/1lMhoWXA0EeZv6tajqXZuEv2sMHdg9zUr?usp=sharing

---

Output

Training Accuracy Graph and Training Loss Graph




```
Summary
----------------------------------------
Learning Rate   Epochs  Test Accuracy
----------------------------------------
0.001       20  0.0000
0.001       50  0.6667
0.001       100 0.0000
0.01        20  0.6667
0.01        50  0.6667
0.01        100 1.0000
0.1     20  1.0000
0.1     50  1.0000
0.1     100 1.0000
```

Backpropogation:



Example:

Test Loss : 0.0785

Test Accuracy : 97.85%
